

Problem Statements

Anuj Udupure (50)

Om Umale (53)

1. TECHDocRag: Intelligent Document Retrieval and Summarization

Problem Statement:

Accessing and understanding technical documentation can be overwhelming due to its large volume and scattered structure. Users often struggle to find accurate, context-specific answers quickly. TECHDocRag aims to solve this by integrating Retrieval-Augmented Generation (RAG) with a language model. The system ingests documents, stores them in a vector database for semantic search, and retrieves relevant chunks to provide concise, citation-backed summaries. This enables efficient knowledge discovery, faster learning, and reliable insights for technical professionals.

2. Risk Assessment Engine for Loan Approval (Mini Credit Engine)

Problem Statement:

Financial institutions face challenges in efficiently evaluating loan applications due to reliance on manual assessments and limited scoring mechanisms. This results in delays and potential biases in decision-making. A lightweight risk assessment engine is needed to simulate credit scoring based on income, transaction history, and credit behavior. By generating a risk score and loan approval recommendation, this system will demonstrate the application of machine learning and explainable AI (XAI) in reducing risks and improving transparency in lending.

3. Chatbot for Banking Services

Problem Statement:

Traditional banking customer service channels are often time-consuming, leading to delays in resolving simple queries such as account balance checks, fund transfers, or loan-related information. There is a need for an intelligent, voice-enabled chatbot that can provide secure, real-time assistance to customers. This system will leverage NLP to understand both text and voice inputs, handle FAQs, and guide users through banking processes, thereby improving accessibility and enhancing customer experience.